

PROFILE SUMMARY

- Senior Technical Artist with 9 years of experience shipping a AAA open-world RPG franchise across character pipeline, materials, and procedural environments, specializing in shader authoring, Houdini procedural pipelines, and artist-facing Python tooling.
- Solid technical background across engine (Unreal Engine 5), shader tools (Unreal Material Editor + HLSL), DCC stack (Maya, Houdini), texturing on Substance Painter, scripting in Python, VFX on Niagara, procedural pipelines with Houdini Engine + HDAs, and target platforms PS5, Xbox Series X|S, and PC, with strong fundamentals in art-first technical thinking, tight artist feedback loops, and a strong respect for performance budgets.
- Deep expertise in shader authoring and material-library design, art pipeline and DCC-to-engine tooling, rigging, skinning, and runtime animation tech, and procedural pipelines with Houdini and HDAs, applying practices such as performance-budget-aware authoring and artist-enablement scripting and editor extensions to deliver ship-quality, perf-clean, and artist-empowering tech-art systems.
- Engaged collaborator working cross-functionally with Character Art, Environment Art, Animation, VFX, Lighting, and Engineering leadership in long-cycle AAA productions with weekly art reviews, contributing to art-pipeline reviews, perf war rooms, and look-dev retrospectives with a pragmatic, artist-empathy-first mindset.
- Senior IC who shares technical excellence and fosters a culture of art-review and capture-review rigor and performance-budget and naming-convention discipline through tooling-driven enablement and shader mentorship, while running weekly tech-art guild and shader-library review sessions and authoring widely adopted tech-art handbooks.

TECHNICAL SKILLS

Engines & Editors:	Unreal Engine 5 (C++ + Blueprints + Python), Unity 6 (C# editor extensions), Godot 4 (cross-engine study), Perforce, GitHub
Shaders & Materials:	Unreal Material Editor + HLSL custom nodes, Unity Shader Graph + HLSL, Amplify, ShaderFX, stylized + PBR material libraries
DCC & Texturing:	Maya (daily), Houdini, Blender, 3ds Max, ZBrush, Substance Painter, Substance Designer, Marmoset Toolbag
Scripting & Tools:	Python (Maya + Unreal + Houdini), MEL, MaxScript, C# (Unity editor), PySide / Qt for tool UIs, Perforce + Git automation
Rigging & Animation Tech:	Maya rigging, control rigs (Unreal Control Rig), blend shapes, IK / FK, facial rigs (ARKit / FaceWare), runtime animation systems
VFX & Particles:	Niagara, Unity VFX Graph, Houdini sims, flipbooks, particle shaders, texture-driven FX, lightning / fire / smoke / water systems
Procedural & Houdini:	Houdini Engine + HDAs, SideFX Labs, PDG, procedural environments / foliage / destruction / terrain, Substance generators
Performance & Look Dev:	Unreal Insights, RenderDoc, PIX, polycount / LOD authoring, texture compression + atlasing, shader-complexity views, lighting + color-grade authoring

EDUCATION

Savannah College of Art and Design **B.F.A. in Interactive Design & Game Development (Technical Art track)**

Savannah, GA • Sep 2013 - Jun 2017

WORK EXPERIENCE

Bethesda Game Studios **Senior Technical Artist**

Rockville, MD • Sep 2020 - Present

- Serve as the tech-art bridge for Bethesda's flagship open-world RPG on the Creation Engine, supporting 60+ artists across 4 disciplines (Character, Environment, VFX, and Animation), partnering with engineering on weekly art reviews and pipeline RFCs.
- Authored the character, foliage, and weather material library in Unreal Material Editor with 120+ HLSL custom nodes and reusable layered functions, cutting per-asset material authoring from 4 hrs to 40 min while holding the art-director look targets.
- Designed the Maya, Substance, and Houdini to Unreal asset pipeline with 14 strict validators and naming-convention enforcement on every DCC handoff, cutting per-batch asset-import errors from 18% to under 2% across the back half of the program.
- Built Python and PySide auto-rig, asset-validator, and Unreal editor extensions for the art teams, adopted by 40+ artists across all four disciplines and pulling the per-asset iteration loop from 12 min down to 90 sec end-to-end on Perforce.
- Designed a modular character rig with face, cloth, and IK layers on Unreal Control Rig and Maya with Python rigging helpers, shipped on 18 hero and 60 background characters with zero rebuilds across the program lifecycle.
- Built the Niagara magic, weather, and combat-feedback library with the VFX artists, shipping 220+ reusable emitters across 6 launch zones and holding the per-frame VFX budget on-target at 14 ms through three rounds of perf review.
- Built the Houdini HDA suite for forests, roads, and destruction with the environment-art team and SideFX Labs, shipping 22 HDAs adopted by 18 env artists on day one and lifting environment-art throughput by 3.2x across the launch zones.

Crystal Dynamics **Technical Artist**

Redwood City, CA • Jul 2017 - Aug 2020

- Drove the perf pass on mesh LODs, texture atlasing, and shader-complexity passes across the studio's flagship action-adventure title, delivering 2.1 ms of draw-time savings per frame with no perceptual loss.
- Partnered with Lighting and Environment Art on lighting templates, post-process volumes, and color-grade LUTs, hitting visual-target sign-off on first review for 8 of 9 cinematic sequences across the 9-sequence vertical slice.
- Mentored 3 mid-career artists through senior tech-art work, supporting 2 promotions to Senior, and authored the team's tech-art handbook with rigging, shader, and perf-budget chapters adopted across 3 art teams.
- Partnered with Character Art, Environment Art, Animation, VFX, Lighting, and Engineering teams across 2 shipped titles and 3 internal tech-demos, authoring 34 tool-design RFCs and shader-library breakdowns that became the team's tech-art baseline and onboarded 6 new artists onto the tooling stack.